

ECE321

Electronic circuits

Lecture (01)- Frequency Response_01

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- ☐ **Basic Concept**
- ☐ **Capacitance Existence**
- ☐ **Typical Frequency Response**
- ☐ **RC Frequency Response**
 - ☐ **Exercise 1**
 - ☐ **Exercise 2**
- ☐ **Gain Bandwidth Product**
- ☐ **Logarithms and Decibel**



- ☐ **Basic Concept**
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Basic Concept (1/2)



- Study impact of frequency/ angular frequency on the amount of voltage gain/current gain or even power gain

Frequency response is a relation between the frequency and gain of the circuit

f : Frequency (Hz)

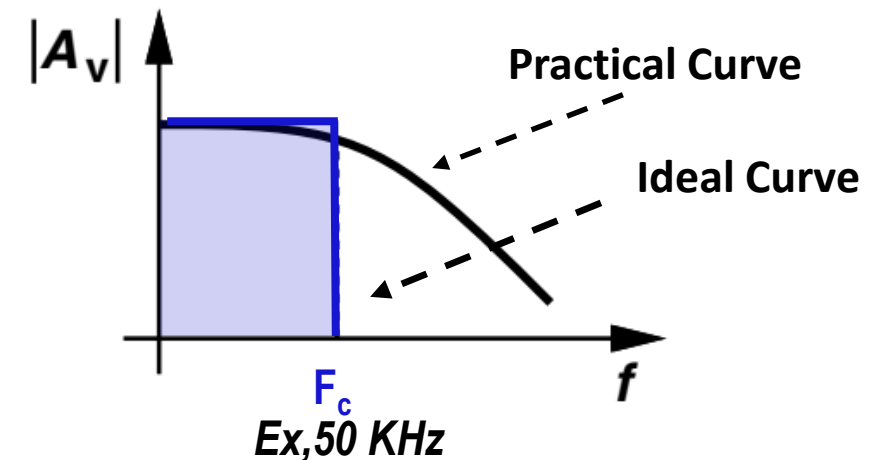
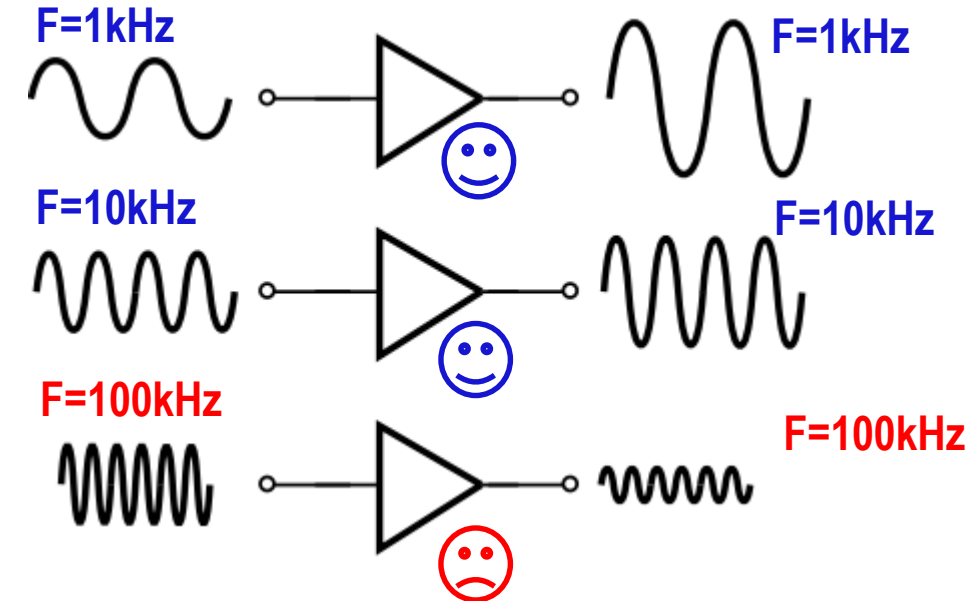
ω : Angular Frequency(rad/s) $\rightarrow \omega = 2\pi f$

A_v : Voltage gain= $\frac{V_o}{V_i}$

A_i : Current gain= $\frac{I_o}{I_i}$

A_p : Power gain= $\frac{p_o}{p_i}$

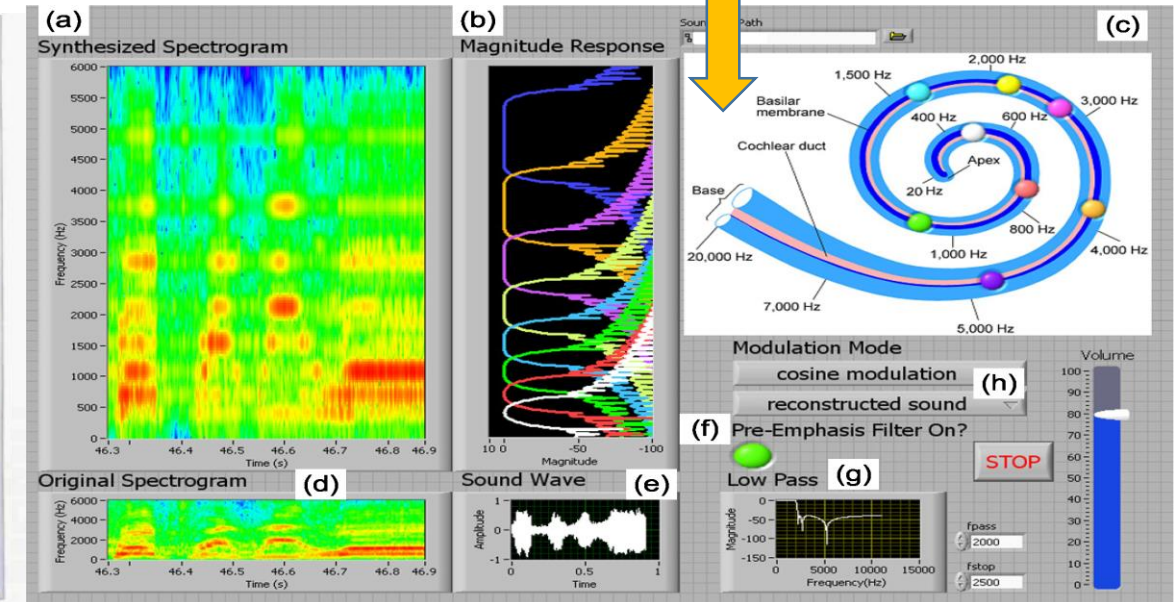
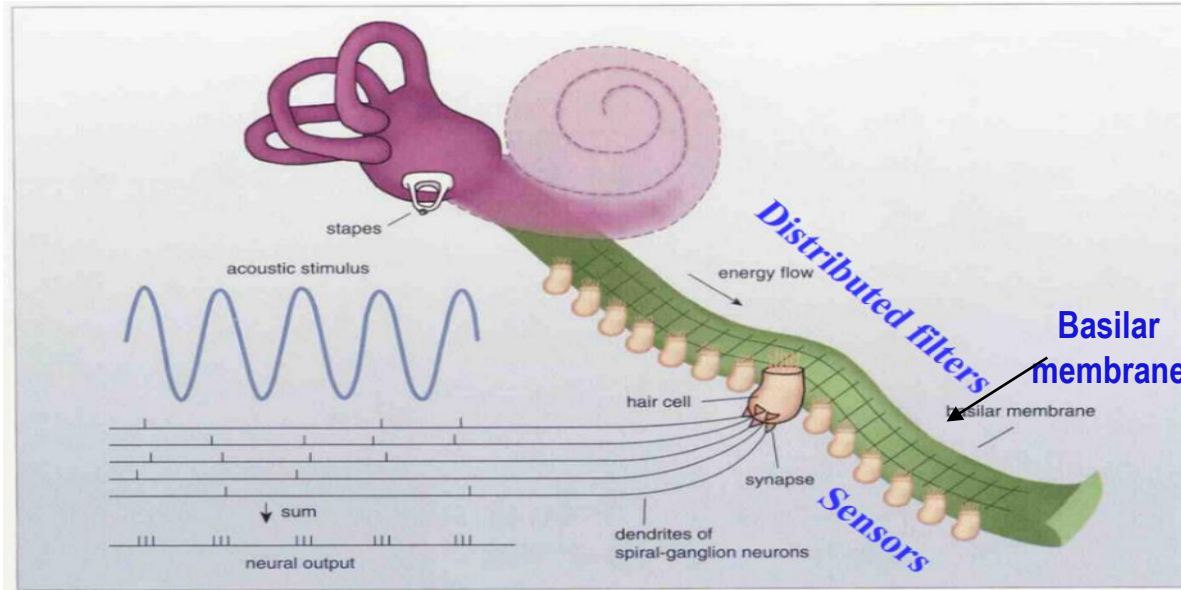
f_c :Cut-off frequency



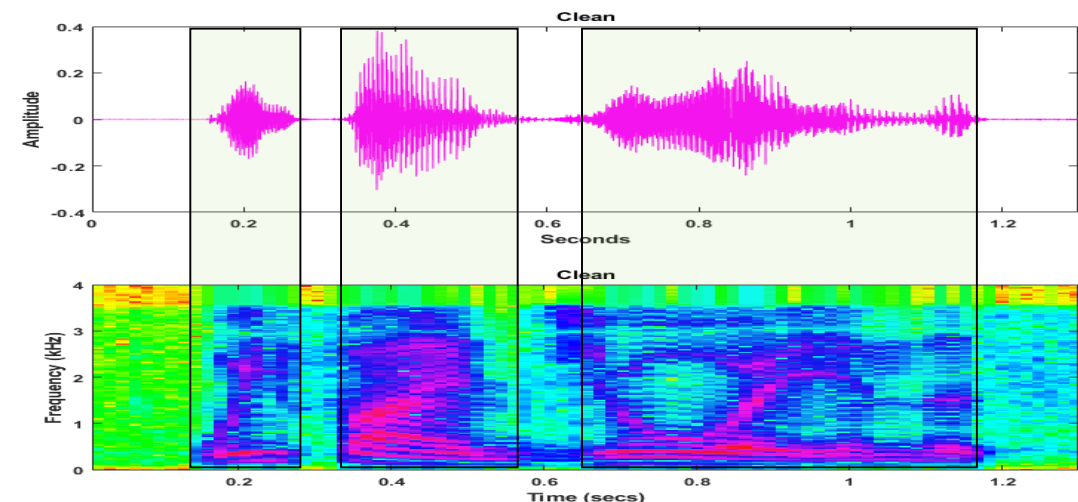
Basic Concept (2/2)



Frequency to place mapping of cochlea



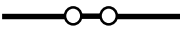
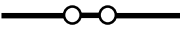
- ❑ The human voice contains frequency components from 20 Hz to 20 kHz.
- ❑ Thus, circuits processing the voice must accommodate this frequency range.



Basic Concept (3/3)



- ❑ Main components that effect significantly the frequency response are capacitor and inductor

Element	Analysis	Equivalent circuit
$\frac{C}{\text{---} \text{---}}$ $X_C = \frac{1}{2\pi f C}$	@ Dc $\rightarrow f=0$ Hz	$X_C = \infty \rightarrow$ Open Circuit 
	@Ac $\rightarrow f=\infty$ Hz	$X_C = 0 \rightarrow$ Short Circuit 
$\frac{L}{\text{---}\text{---}\text{---}}$ $X_L = 2\pi f L$	@Dc $\rightarrow f=0$ Hz	$X_L = 0 \rightarrow$ Short Circuit 
	@Ac $\rightarrow f=\infty$ Hz	$X_L = \infty \rightarrow$ Open Circuit 

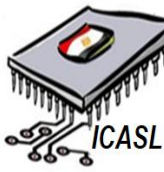
Notes:

- ❑ In DC and AC analysis, we did not consider the frequency impact.
- ❑ So, we need to study the frequency response.   

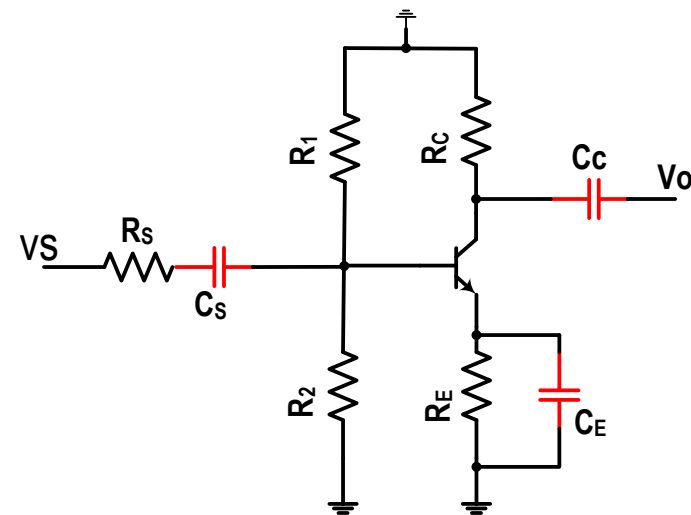


- ☐ Basic Concept
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- ☐ Logarithms and Decibel

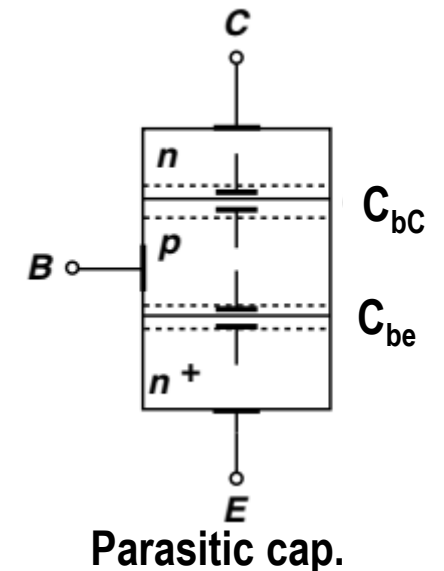
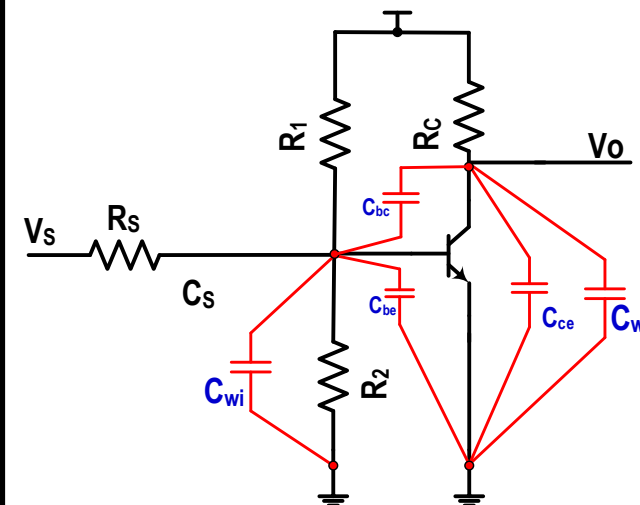
Capacitance Existence



- ❑ Capacitors that are used in your circuit and you **bought** them by your self
 - ❑ $C_s \rightarrow C_i$, $C_c \rightarrow C_o$, and $C_E \rightarrow C_p$
 - ❑ Their effect appears at **low** frequency
 - ❑ Those capacitors will be considered as **short** circuit at high frequency
- ❑ Capacitors that are generated due to **parasitic**, **wiring**, and **soldering**
 - ❑ C_{wi} , C_{wo} , C_{be} , C_{bc} , C_{ce}
 - ❑ Their effect appears only at **high** frequency.



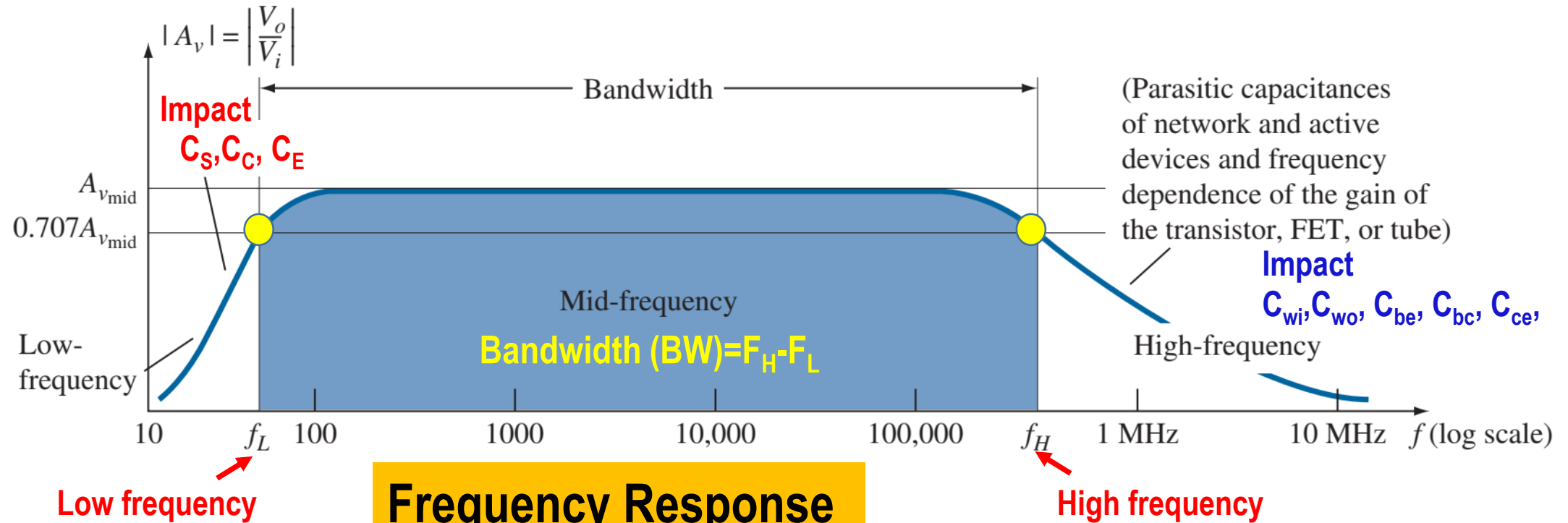
Commercial cap.





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Typical Frequency Response (1/3)



Frequency Response

- F_L and F_H :
- Corner Frequency
 - Cut off frequency
 - Break frequency
 - Half-power frequency
 - -3dB Frequency

$$P_{o,mid} = \frac{V_{o,mid}^2}{R_o}$$

$$P_{o,corner} = \frac{V_{o,corner}^2}{R_o}$$

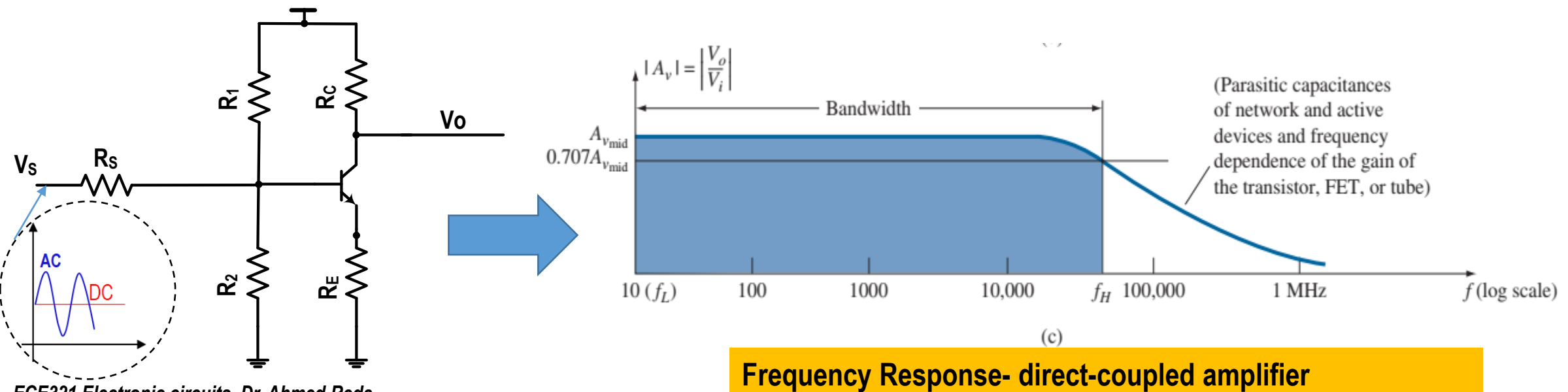
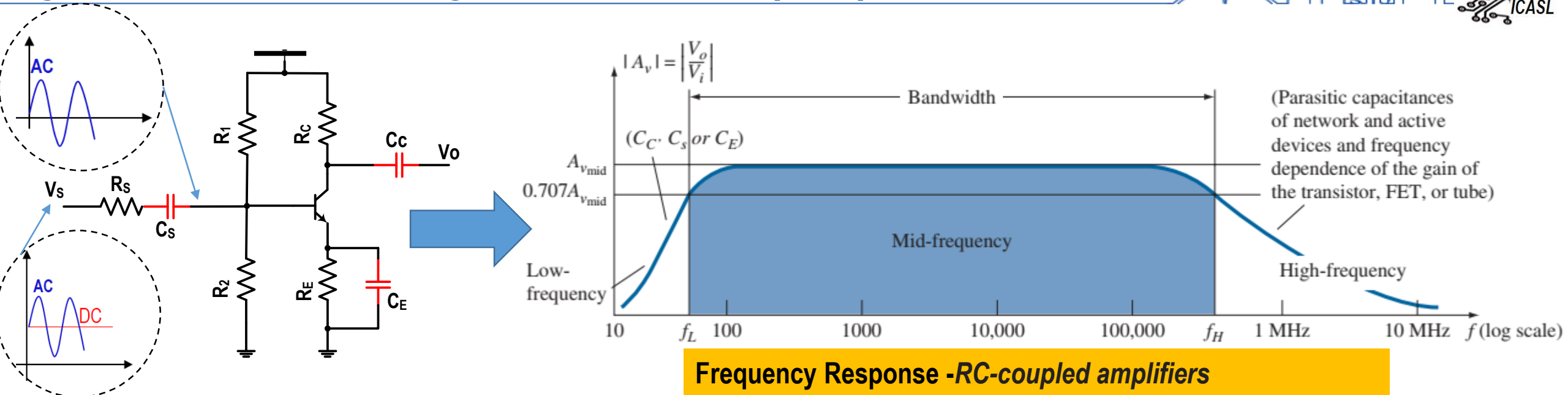
$$= \frac{V_{o,mid}^2}{2 * R_o}$$

Where, $V_{o,corner} = \frac{V_{o,mid}}{\sqrt{2}}$

$P_{o,corner} = \frac{P_{o,mid}}{2}$



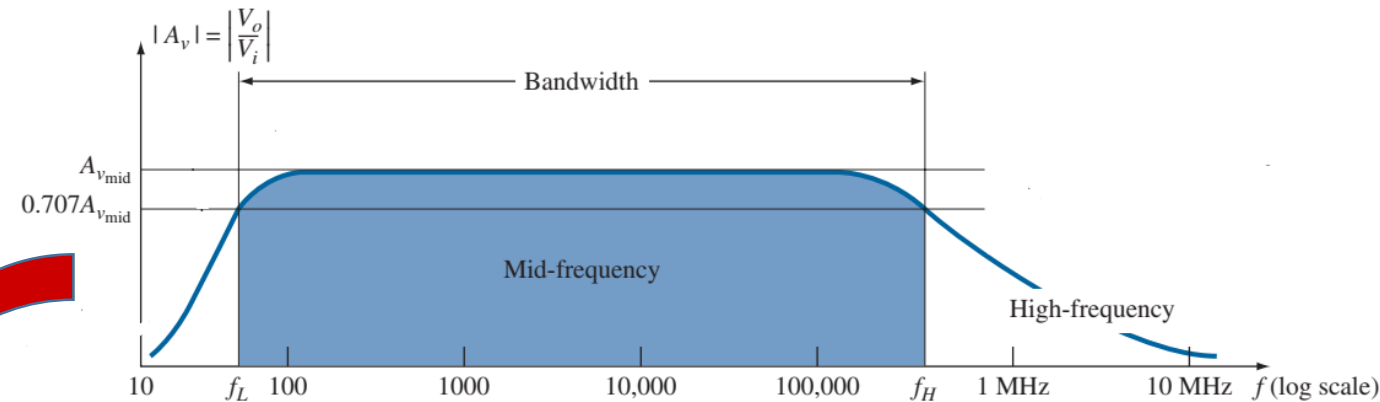
Typical Frequency Response (2/3)



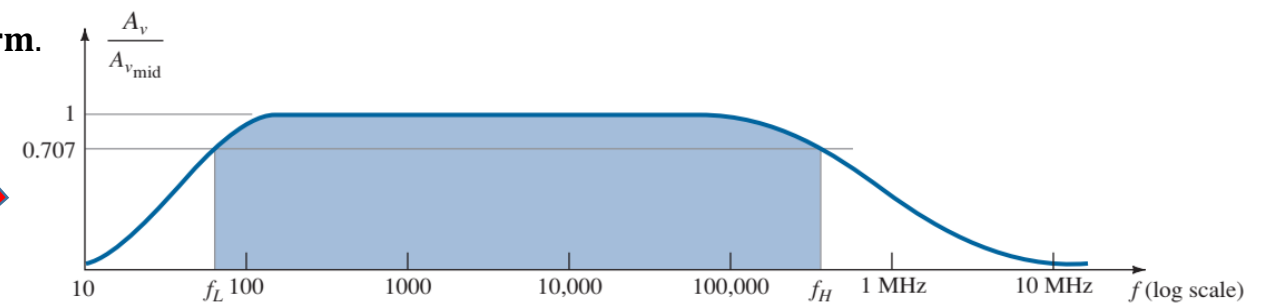
Typical Frequency Response (3/3)



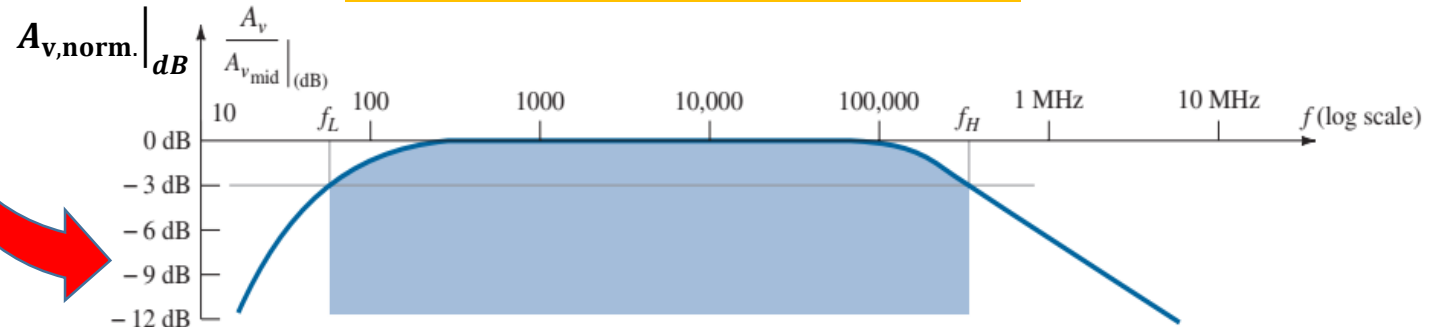
- ❑ The effect of the capacitive elements in an amplifier are **ignored** for the mid-frequency range
- ❑ The band frequencies define a level where the gain or quantity of interest will be **70.7%** of its maximum value .



$A_{v, \text{norm.}}$



Normalized gain versus frequency plot



Decibel plot of the normalized gain versus frequency plot

$$A_{v, \text{norm.}}|_{\text{dB}} = 20 \log(A_{v, \text{norm.}})$$

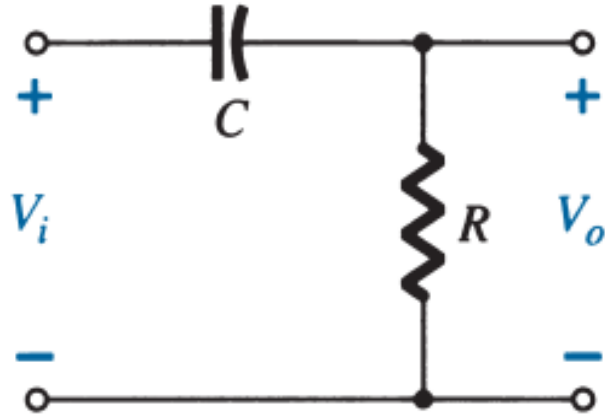


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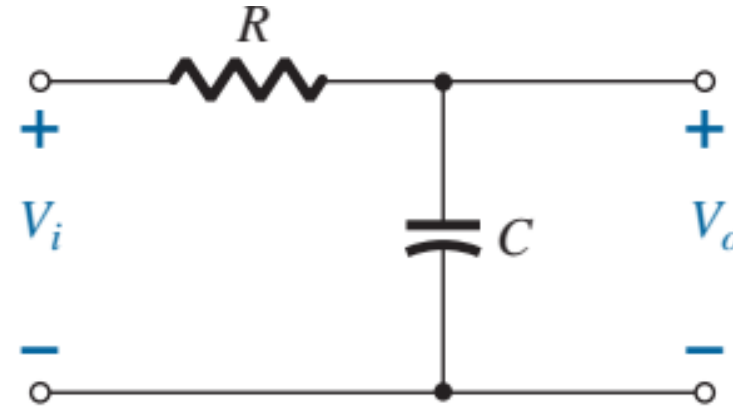
RC Frequency Response



□ In the single-stage **BJT** or *FET* amplifier, **The RC combinations** **formed** by the network capacitors C_C , C_E , and C_S and the network resistive parameters can determine the **cutoff frequencies**.



RC combination that will define a low-cutoff frequency.



RC combination that will define a high-cutoff frequency



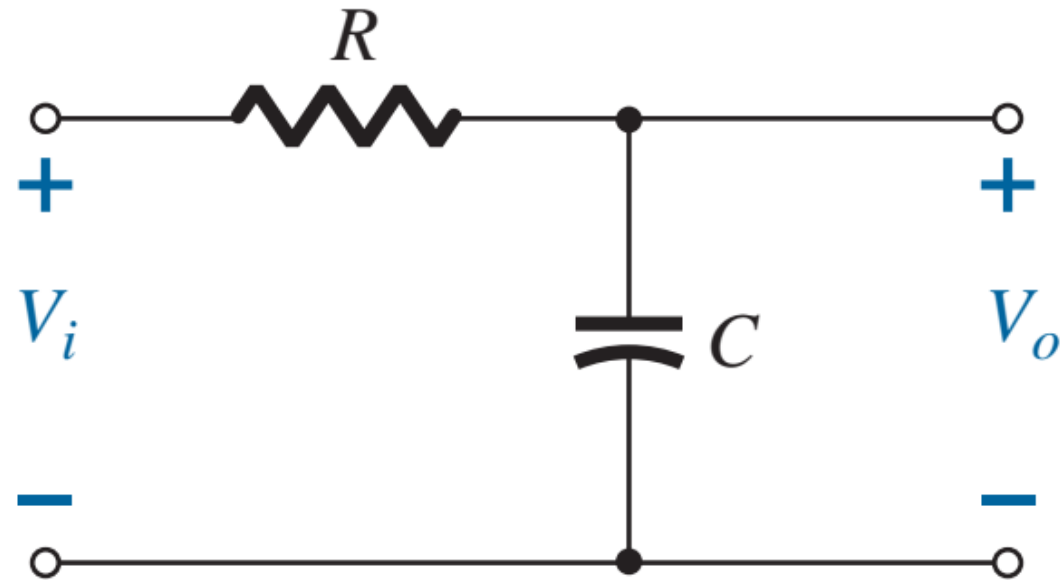
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RC Frequency Response (1/7)



□ **Exercise .1**, for the RC circuit , please determine the next:

1. $A_v(j\omega)$
2. $|A_v(j\omega)|$ & $\angle(A_v(j\omega))$
3. Dc voltage gain=Maximum voltage gain
4. Cut-off frequency= -3dB freq.
5. $|A_v(j\omega)|$ @ $f= 10* f_c$ in dB
6. $|A_v(j\omega)|$ @ $f= 2* f_c$ in dB
7. $|A_v(j\omega)|$ @ $f=\infty$
8. Draw the frequency response
9. Draw the phase response



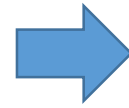
RC Frequency Response (2/7)



Ans.

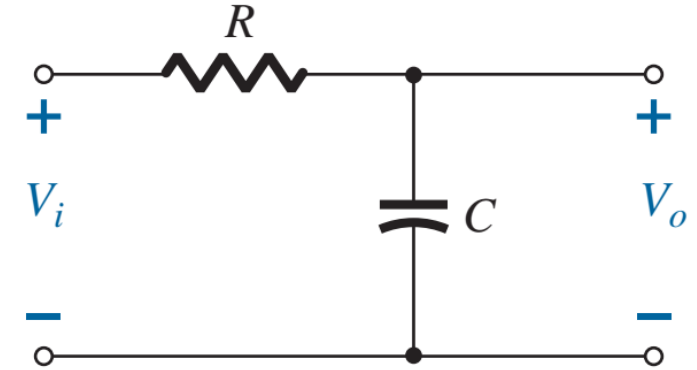
1-To get $A_v(j\omega)$

$$V_o = V_i * \frac{X_c}{R+X_c} = V_i * \frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}} = V_i * \frac{1}{j\omega RC+1}$$



$$A_v(j\omega) = \frac{1}{j\omega RC+1}$$

$$A_v(s) = \frac{1}{sRC+1}$$



2- To get $|A_v(j\omega)|$ and $\phi(A_v(j\omega))$

$$|A_v(j\omega)| = \frac{1}{\sqrt{1+(\omega RC)^2}}$$

$$\phi(A_v(j\omega)) = \phi_{\text{numerator}} - \phi_{\text{denominator}}$$

$$= \tan^{-1}\left(\frac{\text{imag. part}}{\text{Real part}}\right) - \tan^{-1}\left(\frac{\text{imag. part}}{\text{Real part}}\right)$$

$$= \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{\omega RC}{1}\right)$$



$$\phi(A_v(j\omega)) = -\tan^{-1}(\omega RC)$$

RC Frequency Response (3/7)



Ans.

3-To get Dc – Gain

It means that what is the value of voltage gain @ **ZERO** freq. $\rightarrow f=0$ Hz, $w=0$ rad/s, $s=0$

$$|A_v(jw)| = \frac{1}{\sqrt{1+(wRC)^2}} \Rightarrow \boxed{\text{Dc – Gain} = 1}$$

4-To cut-off freq. = -3dB freq.

□ It means that what is the value of freq. when the power = $0.5 \times \text{Power}_{\max}$

□ It means that what is the value of freq. when the voltage gain = $\frac{\text{Voltage gain}_{\max}}{\sqrt{2}} = \frac{\text{Dc-Gain}}{\sqrt{2}}$

$$|A_v(jw_c)| = \frac{\text{Dc-Gain}}{\sqrt{2}}$$
$$\frac{1}{\sqrt{1+(w_cRC)^2}} = \frac{1}{\sqrt{2}}$$

$$1 + (w_cRC)^2 = 2 \Rightarrow w_cRC = 1$$

$$w_c = \frac{1}{RC} \Rightarrow \boxed{f_c = \frac{1}{2\pi RC}}$$

Note:

□ For RC Circuit, We can say that $f_c = \frac{1}{2\pi R_{eq} C_{eq}}$

$$|A_v(jw)| = \frac{1}{\sqrt{1+(wRC)^2}}$$

$$|A_v(jw)| = \frac{1}{\sqrt{1+(\frac{w}{w_c})^2}}$$

$$|A_v| = \frac{1}{\sqrt{1+(\frac{f}{f_c})^2}}$$

$$\phi(A_v(jw)) = -\tan^{-1}(wRC)$$

$$\phi(A_v(jw)) = -\tan^{-1}(\frac{w}{w_c})$$

$$\phi(A_v) = -\tan^{-1}(\frac{f}{f_c})$$



RC Frequency Response (4/7)



Ans.

To $|A_v(j\omega)|$ @ $f = 10 * f_c$ in dB

$$|A_v| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

Since $f \gg f_c \Rightarrow 1 + \left(\frac{f}{f_c}\right)^2 \approx \left(\frac{f}{f_c}\right)^2$

$$|A_v| \cong \frac{f_c}{f}$$

5- $|A_v|$ @ $f = 10 * f_c$

$$|A_v| \cong 0.1$$

$$\begin{aligned} |A_v| &\cong 20 \log 0.1 \\ &\cong -20 \text{ dB} \end{aligned}$$

6- $|A_v|$ @ $f = 2 * f_c$

$$|A_v| \cong 0.5$$

$$\begin{aligned} |A_v| &\cong 20 \log 0.5 \\ &\cong -6 \text{ dB} \end{aligned}$$

Note:

□ Voltage gain will **decreases** by **20 dB** @ $f = 10 * f_c$

□ It means $\rightarrow -20\text{dB/decade}$

□ Voltage gain will **decreases** by **6 dB** at $f = 2 * f_c$

□ It means $\rightarrow -6\text{dB/octave}$

7-To $|A_v|$ @ $f = \infty$

$$|A_v| = 0$$

RC Frequency Response (5/7)



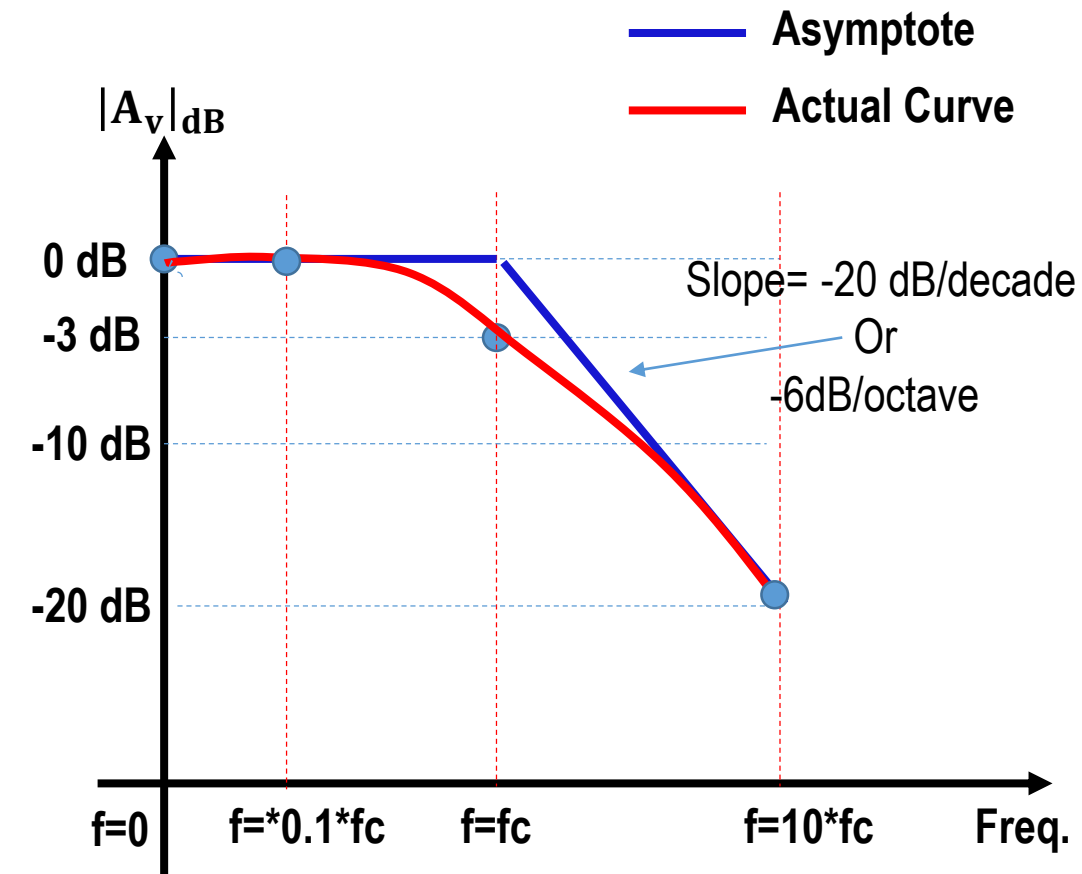
Ans.

8-To draw the frequency response

It is a relation between the magnitude of voltage gain and freq.

$$|A_v| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

f	$ A_v $	$ A_v _{dB}$
0	1	0 dB
$0.1 \cdot f_c$	0.995	$\sim 0 \text{ dB}$ (-0.04 dB)
f_c	$1/\sqrt{2}$	-3dB
$10 \cdot f_c$	$\sim .1$	-20 dB
∞	0	$-\infty$



RC Frequency Response (6/7)



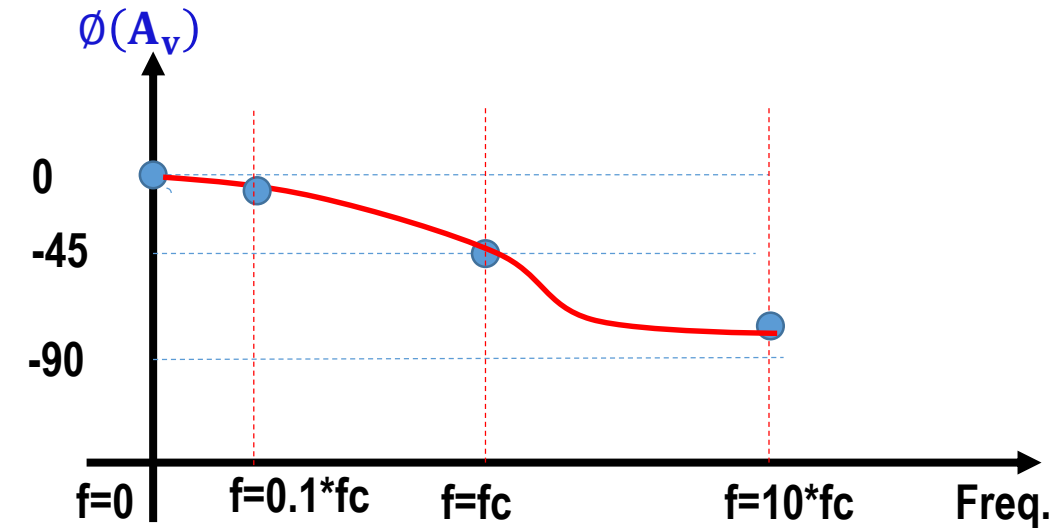
Ans.

9-To draw the Phase Response

It is a relation between the Phase of voltage gain and freq.

$$\phi(A_v) = -\tan^{-1}\left(\frac{f}{f_c}\right)$$

f	$\phi(A_v)$
0	0
$0.1 * f_c$	-5.7
f_c	-45
$10 * f_c$	-84
∞	-90

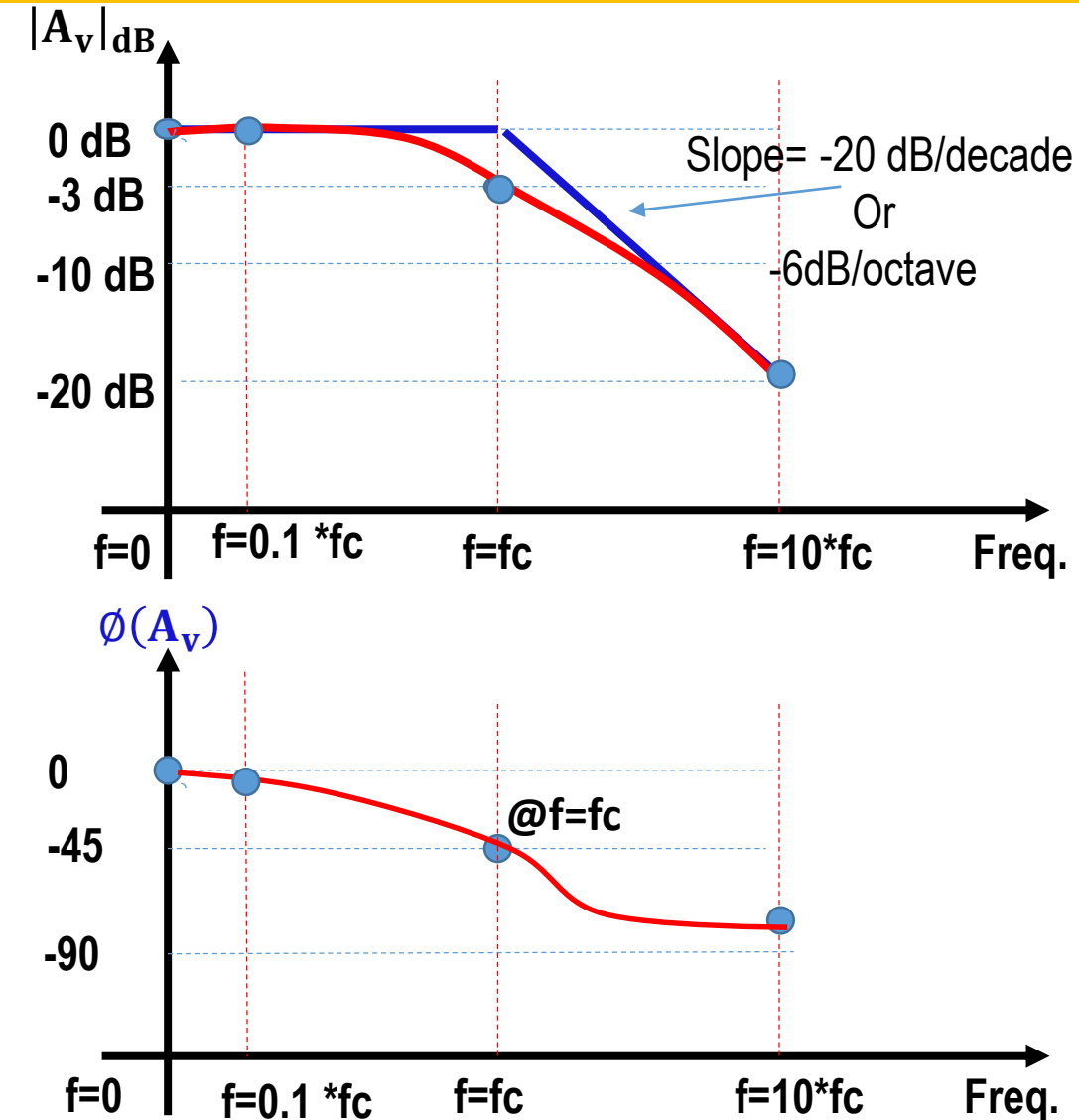


RC Frequency Response (7/7)



Ans.

f	$ A_v $	$ A_v _{dB}$	$\phi(A_v)$
0	1	0 dB	0
f_c	$1/\sqrt{2}$	-3dB	-45
$10^* f_c$	$\sim .1$	-20 dB	-84
∞	$\sim \infty$	$\sim \infty$	-90



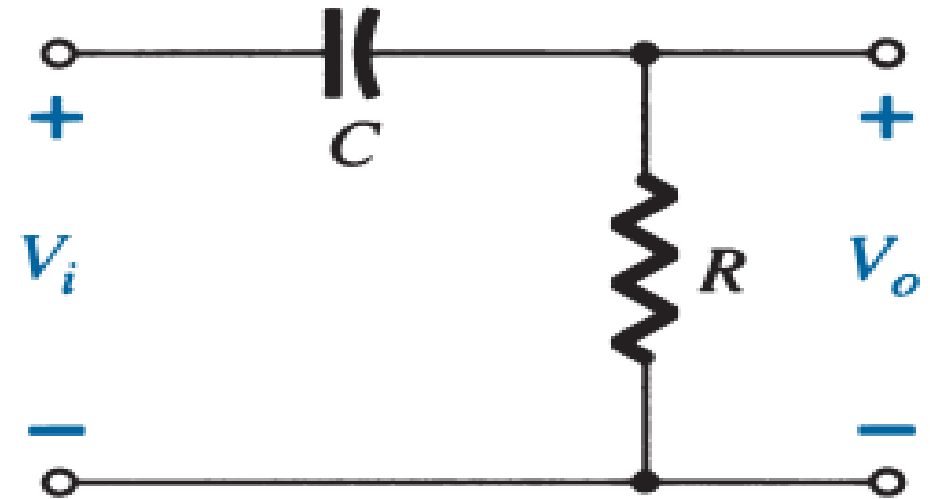
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RC Frequency Response (1/6)



□ Exercise .2, for the RC circuit , please determine the next:

1. $A_v(j\omega)$
2. $|A_v(j\omega)|$ & $\phi(A_v(j\omega))$
3. Mid-band voltage gain= Maximum voltage gain
4. Cut-off Frequency
5. $|A_v(j\omega)|$ @ $f=0.1 f_c$ in dB
6. $|A_v(j\omega)|$ @ $f=0.5 f_c$ in dB
7. $|A_v(j\omega)|$ @ $f=\infty$
8. Draw the frequency response
9. Draw the phase response



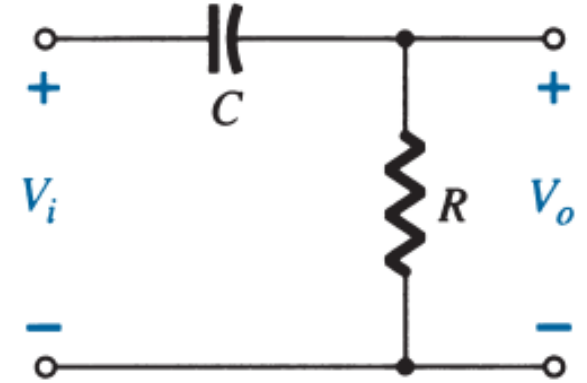
RC Frequency Response (2/6)

Ans.



1-To get $A_v(j\omega)$

$$V_o = V_i * \frac{R}{R+X_c} = V_i * \frac{R}{R+\frac{1}{j\omega}} = V_i * \frac{1}{1-\frac{j}{\omega RC}} \Rightarrow A_v(j\omega) = \frac{1}{1-\frac{j}{\omega RC}}$$



2- To get $|A_v(j\omega)|$ and $\phi(A_v(j\omega))$

$$|A_v(j\omega)| = \frac{1}{\sqrt{1+(\frac{1}{\omega RC})^2}}$$

$$\omega_c = \frac{1}{RC}$$

$$|A_v(j\omega)| = \frac{1}{\sqrt{1+(\frac{\omega_c}{\omega})^2}}$$

$$\omega_c = 2\pi f_c$$

$$|A_v(j\omega)| = \frac{1}{\sqrt{1+(\frac{f_c}{f})^2}}$$

$$\begin{aligned}\phi(A_v(j\omega)) &= \phi_{\text{numerator}} - \phi_{\text{denominator}} \\ &= \tan^{-1}\left(\frac{\text{imag. part}}{\text{Real part}}\right) - \tan^{-1}\left(\frac{\text{imag. part}}{\text{Real part}}\right) \\ &= \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{-1}{\omega RC}\right)\end{aligned}$$

$$\phi(A_v(j\omega)) = +\tan^{-1}\left(\frac{1}{\omega RC}\right)$$

$$\phi(A_v(j\omega)) = +\tan^{-1}\left(\frac{f_c}{f}\right)$$

RC Frequency Response (3/6)



Ans.

3-To get maximum voltage gain =mid-band gain

$$|A_v| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}} \quad \text{@ } f_c = \infty$$

, we can get max. voltage gain
 $|A_v| = 1$

4-To cut-off freq. = -3dB freq.

□ For RC Circuit , We can say that $f_c = \frac{1}{2\pi R_{eq} C_{eq}}$

$$f_c = \frac{1}{2\pi RC}$$

To $|A_v(j\omega)|$ @ $f = 0.1 * f_c$ in dB

$$|A_v| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}} \quad \text{Since } f \ll f_c \Rightarrow |A_v| \approx \frac{f}{f_c}$$
$$1 + \left(\frac{f}{f_c}\right)^2 \approx \left(\frac{f}{f_c}\right)^2$$

5- $|A_v|$ @ $f = 0.1 * f_c$

$$|A_v| \approx 0.1 \Rightarrow |A_v| \approx 20 \log 0.1 \approx -20 \text{ dB}$$

6- $|A_v|$ @ $f = 0.5 * f_c$

$$|A_v| \approx 0.5 \Rightarrow |A_v| \approx 20 \log 0.5 \approx -6 \text{ dB}$$

Note:

- Voltage gain will **decreases** by 20 dB at $f = 0.1 f_c$
 - It means $\rightarrow -20\text{dB/decade}$
- Voltage gain will **decreases** by 6dB at $f = 0.5 * f_c$
 - It means $\rightarrow -6\text{dB/octave}$

7-To $|A_v|$ @ $f = \infty$

$$|A_v| = 1$$

RC Frequency Response (4/6)



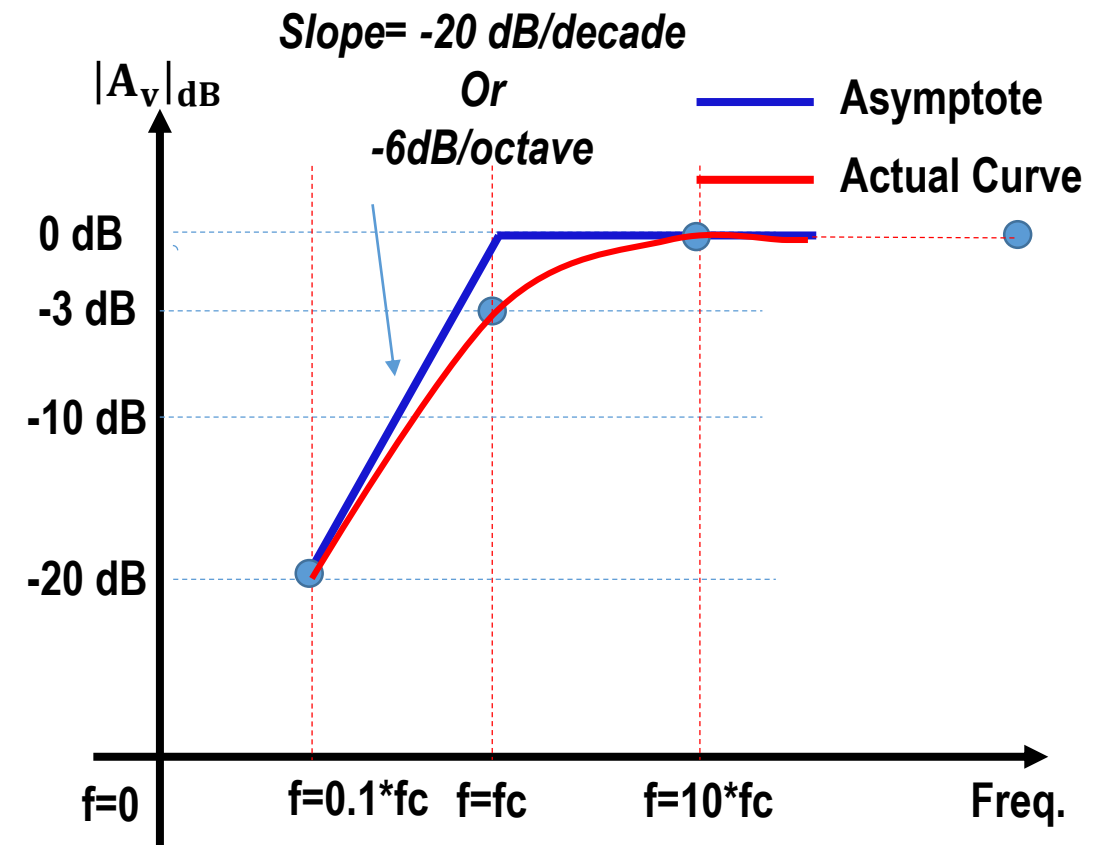
Ans.

8-To draw the frequency response

It is a relation between the Magnitude of voltage gain and freq.

$$|A_v| = \frac{1}{\sqrt{1 + \left(\frac{f_c}{f}\right)^2}}$$

f	$ A_v $	$ A_v _{dB}$
0	0	$-\infty$ dB
$0.1 * f_c$	~ 0.1	-20 dB
f_c	$1/\sqrt{2}$	-3dB
$10 * f_c$	0.995	~ 0 dB
∞	1	0dB



RC Frequency Response (5/6)



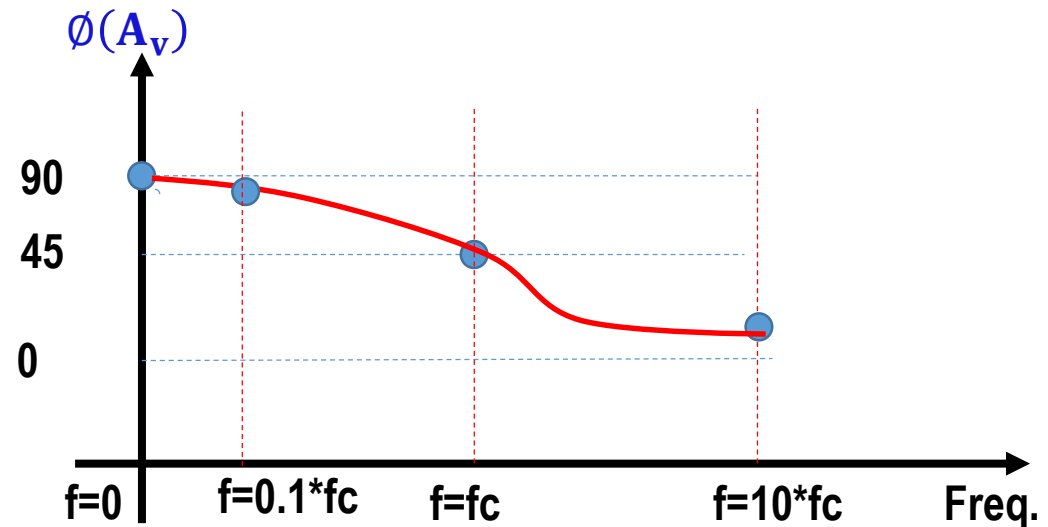
Ans.

9-To draw the Phase Response

It is a relation between the Phase of voltage gain and freq.

$$\phi(A_v) = + \tan^{-1}\left(\frac{f_c}{f}\right)$$

f	$\phi(A_v)$
0	90
$0.1 * f_c$	84.2
f_c	45
$10 * f_c$	5.7
∞	0

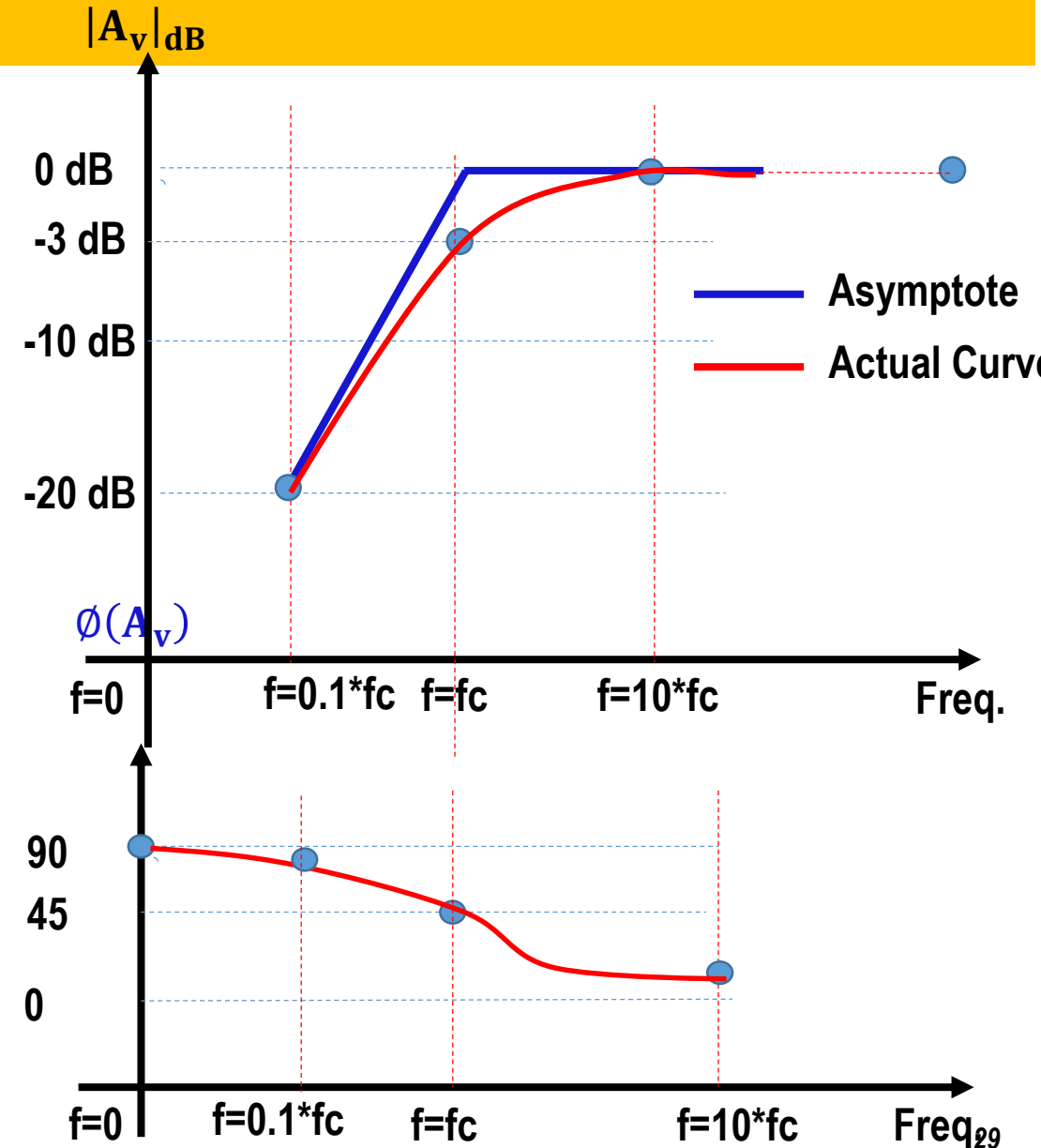


RC Frequency Response (6/6)



Ans.

f	$ A_v $	$ A_v _{dB}$	$\phi(A_v)$
0	0	$-\infty$ dB	90
$0.1 * f_c$	~ 0.1	-20 dB	84.2
f_c	$1/\sqrt{2}$	-3dB	45
$10 * f_c$	0.995	~ 0 dB	5.7
∞	1	0dB	0



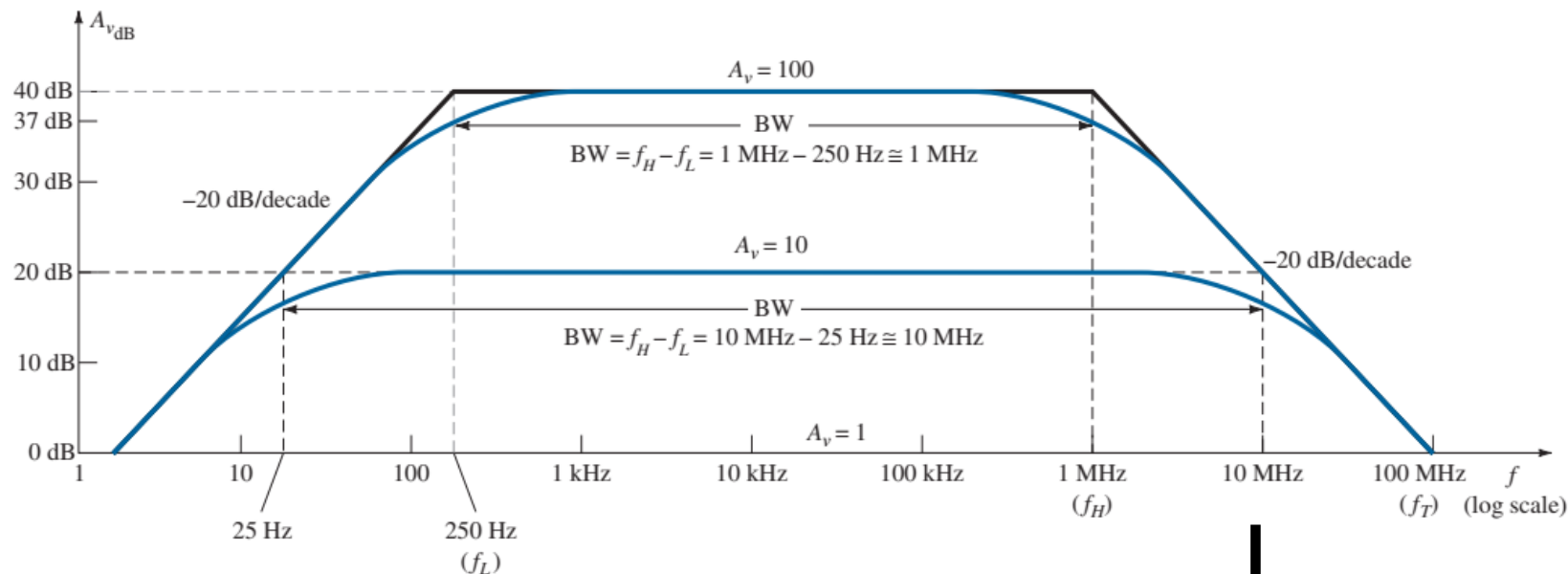


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Gain Bandwidth Product (1/3)



- ❑ There is a **Figure of Merit** applied to amplifiers called the **Gain-Bandwidth Product (GBP)** that is commonly used to initiate the design process of an amplifier .
- ❑ It provides important information about the relationship between the gain of the amplifier and the expected **operating frequency range**.



Ex.1, $GBP = (100)(1 \text{ MHz} - 250 \text{ Hz}) \sim 100 \text{ MHz}$

Ex.2, $GBP = (10)(10 \text{ MHz} - 25 \text{ Hz}) \sim 100 \text{ MHz}$

In fact, at any level of gain the product of the two remains a constant

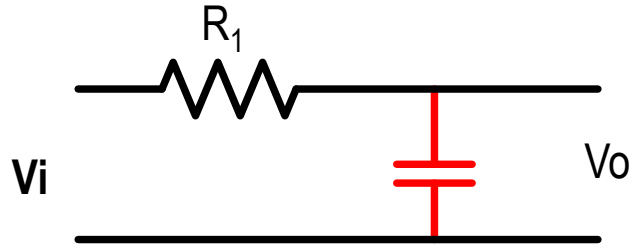
$$GBP = A_{vmid} * Bw$$

@ $A_v = 1$ or $A_v(\text{dB}) = 0$ bandwidth is defined as f_T
the frequency f_T is called the unity-gain frequency and is always equal to the product of the midband gain of an amplifier and the bandwidth at any level of gain.

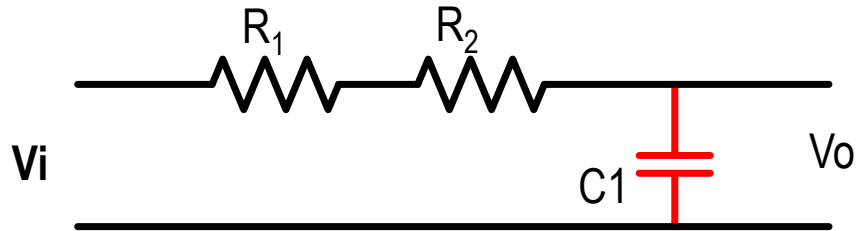
$$f_t = GBP$$

Gain Bandwidth Product (2/3)

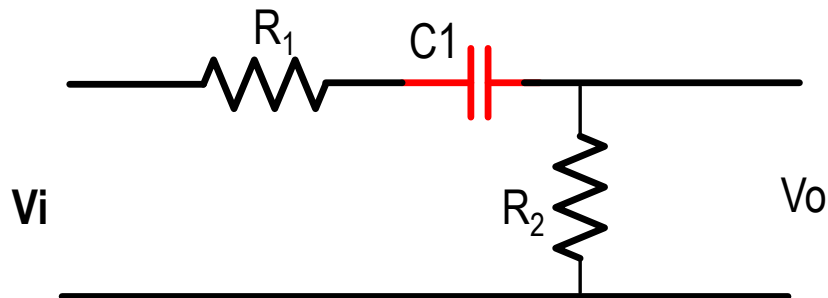
□ Cut-off frequency = break frequency = corner frequency



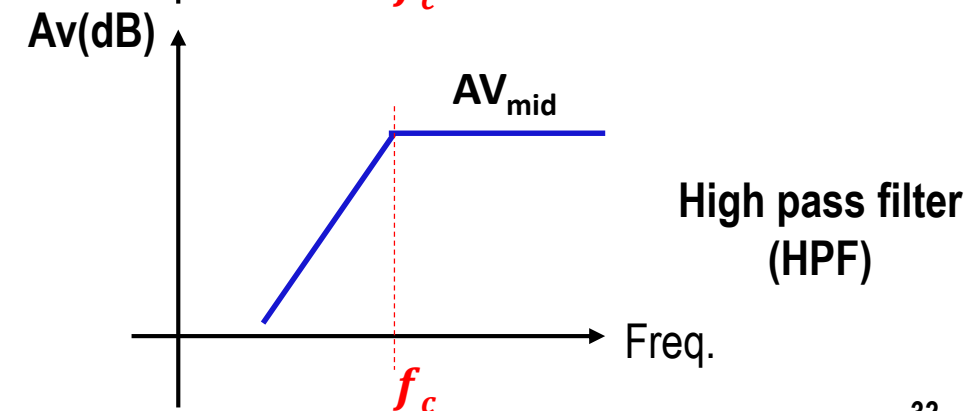
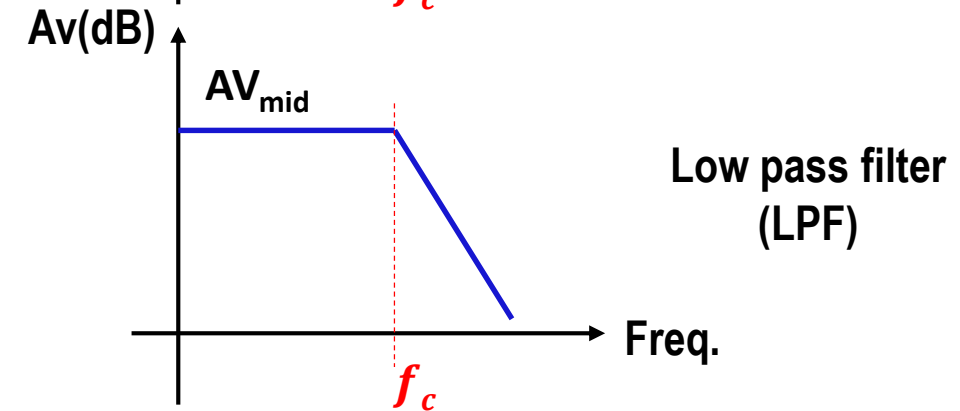
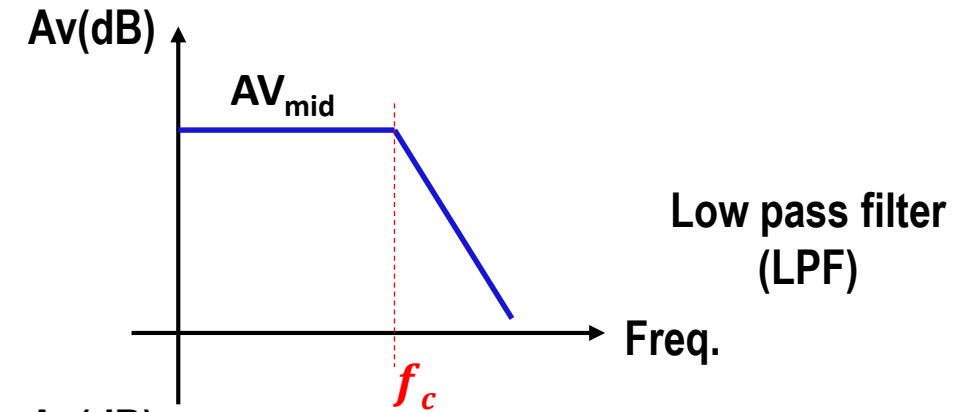
$$f_c = \frac{1}{2\pi R_1 C_1}$$



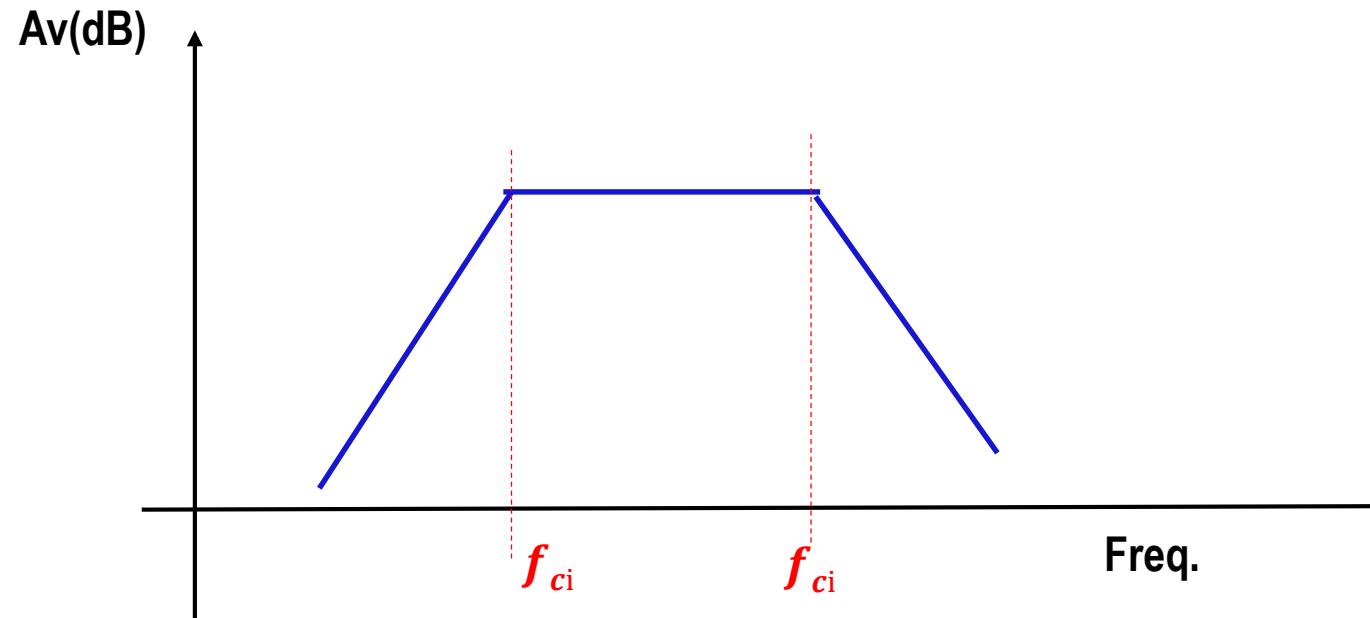
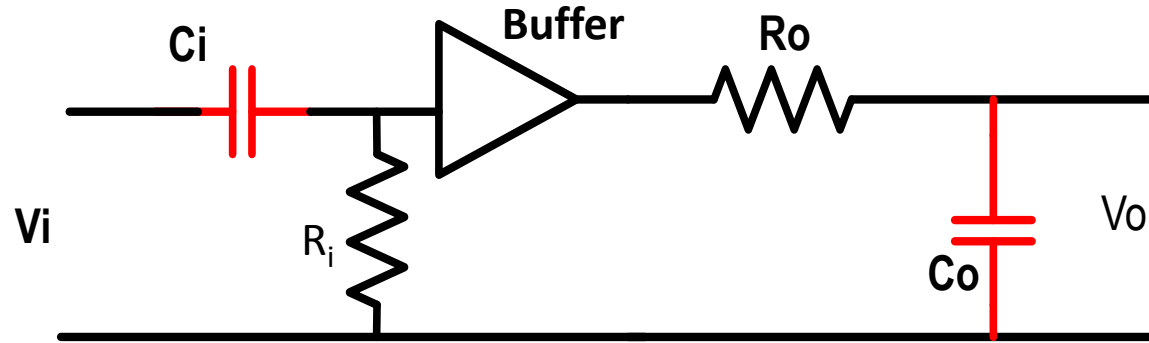
$$f_c = \frac{1}{2\pi (R_1 + R_2) C_1}$$



$$f_c = \frac{1}{2\pi (R_1 + R_2) C_1}$$



Gain Bandwidth Product (3/3)



$$f_{ci} = \frac{1}{2\pi R_i C_i}$$

$$f_{co} = \frac{1}{2\pi R_o C_o}$$

$$BW = f_{ci} - f_{co}$$



- ☐ **Basic Concept**
- ☐ **Capacitance Existence**
- ☐ **Typical Frequency Response**
- ☐ **RC Frequency Response**
 - ☐ **Exercise 1**
 - ☐ **Exercise 2**
- ☐ **Gain Bandwidth Product**
- ☐ **Logarithms and Decibel**

Logarithms and Decibel (1/2)



- Logarithms taken to the base 10 are referred to as *common logarithms*, whereas logarithms taken to the base e are referred to as *natural logarithms*.

$$\text{Common logarithm: } x = \log_{10} a$$

$$\text{Natural logarithm: } y = \log_e a$$

$$\log_{10} 1 = 0$$

$$\log_{10} \frac{a}{b} = \log_{10} a - \log_{10} b$$

$$\log_{10} ab = \log_{10} a + \log_{10} b$$

$$\log_{10} \frac{1}{b} = -\log_{10} b$$

- The terminal rating of electronic communication equipment (amplifiers, microphones, etc.) is commonly in decibels

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1}$$

Output power

Reference/input power

dB

$$G_{\text{dBm}} = 10 \log_{10} \frac{P_2}{1 \text{ mW}} \Big|_{600 \Omega} \quad \text{dBm}$$

$$G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} \quad \text{dB}$$

Logarithms and Decibel (2/2)



❑ The input power to a device is 10,000 W at a voltage of 1000 V. The output power is 500 W and the output impedance is 20 Ω .

- Find the power gain in decibels.
- Find the voltage gain in decibels.

Ans.

$$\begin{aligned} \text{a. } G_{\text{dB}} &= 10 \log_{10} \frac{P_o}{P_i} = 10 \log_{10} \frac{500 \text{ W}}{10 \text{ kW}} \\ &= 10 \log_{10} \frac{1}{20} = -10 \log_{10} 20 \end{aligned}$$

$$= -10(1.301) = \mathbf{-13.01 \text{ dB}}$$

$$\begin{aligned} \text{b. } G_v &= 20 \log_{10} \frac{V_o}{V_i} = 20 \log_{10} \frac{\sqrt{PR}}{1000} \\ &= 20 \log_{10} \frac{\sqrt{(500 \text{ W})(20 \Omega)}}{1000 \text{ V}} \end{aligned}$$

$$= 20 \log_{10} \frac{100}{1000} = 20 \log_{10} \frac{1}{10}$$

$$= -20 \log_{10} 10 = \mathbf{-20 \text{ dB}}$$

❑ An amplifier rated at 40-W output is connected to a 10- Ω speaker.

- Calculate the input power required for full power output if the power gain is 25 dB.
- Calculate the input voltage for rated output if the amplifier voltage gain is 40 dB.

Ans.

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} \text{ dB}$$

$$\text{a. } 25 = 10 \log_{10} \frac{40 \text{ W}}{P_i}$$

$$P_i = \frac{40 \text{ W}}{\text{antilog}(2.5)} = \frac{40 \text{ W}}{3.16 \times 10^2} = \frac{40 \text{ W}}{316} \cong \mathbf{126.5 \text{ mW}}$$

$$\begin{aligned} \text{b. } G_v &= 20 \log_{10} \frac{V_o}{V_i} \\ 40 &= 20 \log_{10} \frac{V_o}{V_i} \end{aligned}$$

$$\frac{V_o}{V_i} = \text{antilog } 2 = 100$$

$$V_o = \sqrt{PR} = \sqrt{(40 \text{ W})(10 \Omega)} = 20 \text{ V}$$

$$V_i = \frac{V_o}{100} = \frac{20 \text{ V}}{100} = 0.2 \text{ V} = \mathbf{200 \text{ mV}}$$

Thanks!